



CHAPTER 5

RESULTS AND DISCUSSION

This chapter evaluates the completed Smart Attendance System as an integrated hardware-software platform. The analysis covers the ESP32-S3 classroom controller, OV2640 image capture, buzzer and LED feedback, PCB and enclosure assembly, secure role-based access, automatic session processing, face attendance, responsive interfaces, weighted calculations, and university-style exports. Reported results are based on implemented behavior and repository verification rather than unsupported estimates of recognition accuracy.

5.1. Analysis of the Complete Hardware-Software System

The implementation connects three user roles and two attendance capture modes to one authoritative backend. Administrators configure academic data, timetables, hardware devices, settings, and reports. Teachers manage assigned courses and attendance records. Students register a face and verify attendance with a mobile camera. The ESP32-S3 and OV2640 provide an alternative fixed classroom capture point.

The integration maintains a clear division of responsibility. Angular manages presentation, Spring Boot applies identity and academic rules, PostgreSQL preserves records, FastAPI handles face embeddings and similarity, and the ESP32-S3 manages camera capture and local feedback. Recognition failure, invalid device authentication, missing enrollment, inactive sessions, and duplicate scans produce controlled non-recording results.

5.1.1. Hardware Integration Results

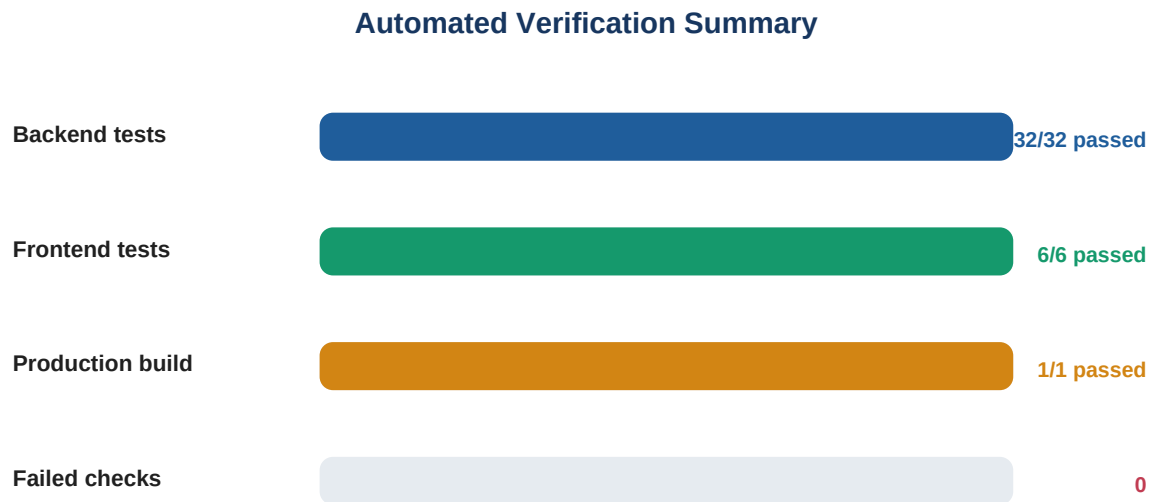
Component	Integrated function	System contribution
ESP32-S3	Controller and Wi-Fi interface	Coordinates capture, API communication, and output control
OV2640 camera	Face image source	Provides the JPEG submitted for candidate identification
Buzzer	Audible status output	Provides immediate local feedback without opening the portal
Green and red LEDs	Visual status outputs	Distinguish successful, rejected, or error states
PCB board	Permanent assembly base	Organizes component interconnections
Junction box	Protective enclosure	Houses the classroom prototype and exposes the camera and indicators

Table 5.1 Hardware integration of the Smart Attendance prototype

The selected component set supports the complete local workflow: image acquisition, authenticated network submission, and visible or audible feedback. Formal bench measurements of capture latency, current draw, indicator timing, enclosure temperature, and long-duration stability remain part of the deployment validation work.

5.2. Functional and Verification Results

Automated verification was used to evaluate the main software behaviors. Existing backend Surefire reports contain 32 passing tests across authentication, academic management, CORS, dashboards, reporting, face-service integration, hardware attendance, notification delivery, and export formatting. The current Angular test run contains six passing tests covering the application shell and optimized authentication and navigation behavior. The Angular production build also completes successfully.



Evidence: repository test reports and the current frontend test/build run.

Figure 5.1 Automated verification results for the current project

Area	Verified result
Authentication	Login, access, refresh-token rotation, logout, replay protection, and role rejection
Academic management	Department, course, assignment, enrollment, timetable, and session consistency
Face attendance	Registration, verification, failed match, duplicate prevention, and FastAPI multipart contract
ESP32-S3 workflow	Managed device registration, secret protection, active-session discovery, identification, and duplicate response
Reports	Readable PDF and Excel output, weighted register, natural roll-number ordering, and dynamic cohort matrix
Responsive frontend	Role-aware pages, mobile attendance cards, loading and error states, and production compilation

Table 5.2 Summary of verified functional outcomes

5.2.1. Attendance Calculation Results

The weighted calculation was verified with four sessions and three roll calls per session. This produces 12 eligible roll calls. Attendance at three sessions contributes nine present roll calls and three absent roll calls, giving $9 / 12 \times 100 = 75.00$ percent. The JSON response, dashboard, PDF, and Excel register use the same totals.

Session date	Roll calls	Recorded state	Present units	Absent units
03/08	3	Present	3	0
10/08	3	Present	3	0
17/08	3	Present	3	0
24/08	3	Absent	0	3
Total	12	-	9	3

Table 5.3 Verified weighted roll-call example

Overall student attendance is calculated across each student's enrolled courses. Tests cover cohorts with different course counts, ensure that non-enrolled course cells remain blank, and order roll numbers such as VMC-2 and VMC-11 naturally rather than lexicographically.

5.2.2. Attendance Decision Results

Condition	System result	Record created
Eligible face matched during active session	ATTENDANCE_VERIFIED	Yes
Same student scans again	ALREADY_VERIFIED	No
Face does not meet threshold	NOT_MATCHED	No
No single device-bound active session	Conflict response	No
Student is not an eligible candidate	Validation failure	No
Device identifier or secret is invalid	Authentication failure	No

Table 5.4 Attendance decision outcomes

5.3. Smart Attendance Operating Modes

5.3.1. ESP32-S3 Automatic Mode

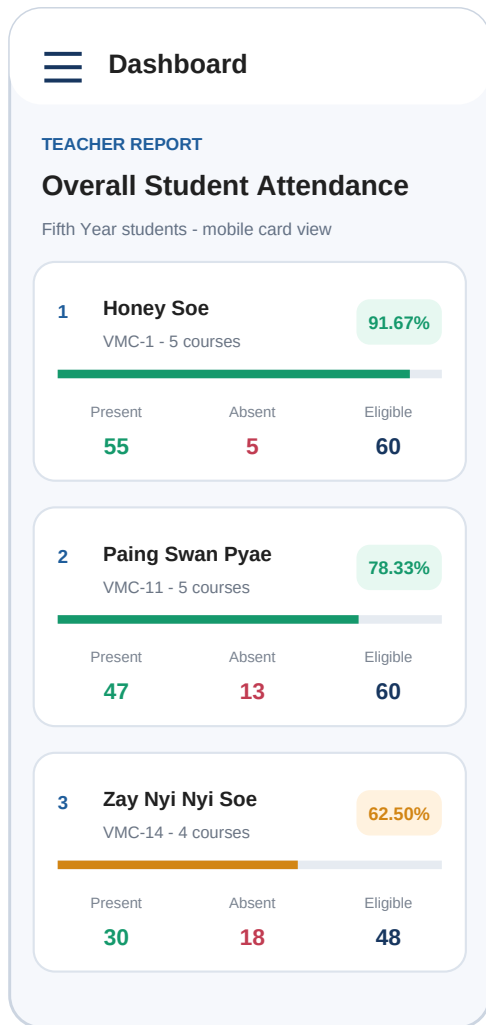
In automatic classroom mode, the ESP32-S3 authenticates with its registered device identifier and secret. Its room or course binding resolves one active session. The OV2640 captures a face image and the firmware submits it for identification. The green and red LEDs and buzzer provide local feedback after the server response. Because academic values remain on the server, the same assembled device can continue operating after timetable changes.

5.3.2. Student Mobile Camera Mode

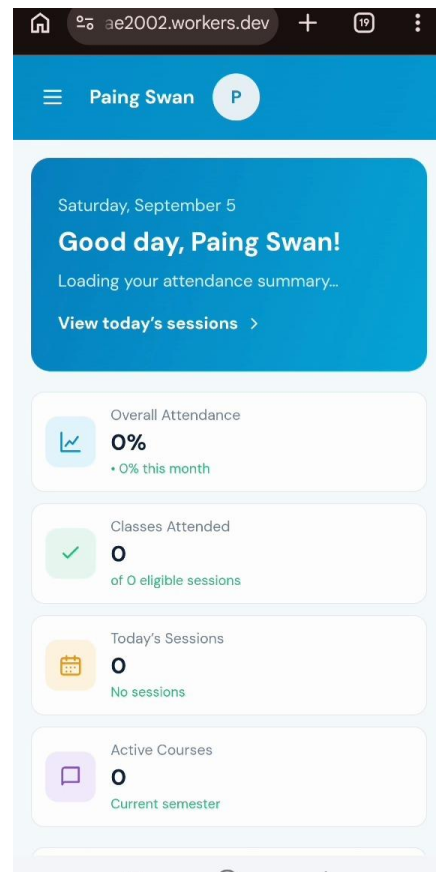
In mobile mode, an authenticated student opens an active session, captures an image, and submits it for verification. The backend derives the student from the JWT and checks enrollment, time window, face registration, and previous attendance. The interface provides permission, capture, verification, success, and retry states for phone use.

5.3.3. Teacher and Administrator Management Mode

Teachers use course rosters, attendance sessions, records, study-year filtering, overall percentages, and Excel downloads. Administrators configure the academic structure, ESP32-S3 device binding, thresholds, audit records, and reports. Wide registers use table layouts on desktop, while mobile student rows use stacked cards so the attendance percentage remains visible.



Teacher overall attendance cards on a mobile viewport



Student dashboard after authenticated login

Figure 5.2 Responsive teacher and student portal layouts

5.3.4. Report and Export Results

The reporting module produces screen-oriented and print-oriented results. The course register follows the university attendance layout with one row per student, date and roll-call columns, present and absent totals, and a percentage. The teacher's overall workbook creates one percentage column per course, calculates each student's overall value with spreadsheet formulas, leaves non-enrolled course cells blank, and provides remark and signature areas.

No.	Roll No.	Student	IA-501	MCE-52018	Control	Overall	Remark
1	VMC-1	Honey Soe	91.67%	88.33%	95.00%	91.67%	Good
2	VMC-2	Hsue Ei Hlaing	45.00%	52.50%	-	48.75%	Review
3	VMC-11	Paing Swan Pyae	78.33%	75.00%	81.67%	78.33%	Good
4	VMC-14	Zay Nyi Nyi Soe	62.50%	60.00%	-	61.25%	Monitor

Table 5.5 Representative teacher cohort matrix layout with illustrative values

Exporter tests reopen generated files and verify structure, readable content, course columns, formulas, print layout, and natural ordering. Download operations are audited by the backend. The frontend sends the selected study year, period, reference date, and search filter to the complete, non-paginated Excel export.

5.4. Discussion

The results demonstrate that the project coordinates hardware, academic data, timetable rules, face recognition, attendance weighting, responsive presentation, and report generation through one consistent server-side model. Candidate-scoped identification reduces unnecessary comparisons, while uniqueness constraints and transaction boundaries protect against repeated scans and scheduler duplication.

The corrected hardware design separates the controller, camera, local feedback, assembly board, and enclosure into clear responsibilities. This makes the classroom unit easier to assemble and maintain than a design that assumes peripherals not present in the prototype. The junction box protects the electronics, while the LEDs and buzzer give users immediate feedback without requiring a browser screen.

5.4. Discussion continued

The system still depends on stable network connectivity among the ESP32-S3, Spring Boot, and FastAPI. Recognition quality depends on lighting, camera position, image clarity, threshold calibration, and registered face data. The current recognition contract must not be treated as production-grade anti-spoofing until a genuine liveness model is validated. Hardware heartbeat monitoring, formal power testing, enclosure testing, and controlled performance measurement are also required before institution-wide deployment.

Recommended future work includes a trained liveness and anti-spoofing pipeline, an offline device queue with signed timestamps, hardware heartbeat and health history, controlled tests under documented lighting and participant conditions, expanded end-to-end tests, accessibility testing, centralized observability, and a tested disaster-recovery procedure.

5.5. Summary

The Smart Attendance System integrates the actual ESP32-S3 hardware unit with an OV2640 camera, buzzer, green and red LEDs, PCB board, and junction box, together with Angular, Spring Boot, PostgreSQL, and FastAPI. Verified software results cover authentication, academic management, automatic sessions, candidate-scoped identification, duplicate prevention, weighted attendance, responsive dashboards, and PDF and Excel reporting. Thirty-two backend tests and six frontend tests pass, and the production frontend build completes. Formal hardware and field-performance validation remains necessary before production deployment.